

1. Employee & Department Reports

Report 1: Employee ID, Full Name, Department Name (Employees with departments only)

Worksheet

Query Builder

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4

SELECT

e.employee_id, e.first_name || ' ' || e.last_name AS employee_name, d.department_name

FROM

employees e

INNER JOIN

departments d

ON

e.department_id = d.department_id;

Query Result x

SQL

| Fetched 100 rows in 0.294 seconds

	EMPLOYEE_ID	EMPLOYEE_NAME	DEPARTMENT_NAME
1	200	Jennifer Whalen	Administration
2	201	Michael Hartstein	Marketing
3	202	Pat Fay	Marketing
4	114	Den Raphaely	Purchasing
5	115	Alexander Khoo	Purchasing
6	116	Shelli Baida	Purchasing
7	117	Sigal Tobias	Purchasing
8	118	Guy Himuro	Purchasing
9	119	Karen Colmenares	Purchasing
10	203	Susan Mavris	Human Resources
11	120	Matthew Weiss	Shipping
12	121	Adam Fripp	Shipping
13	122	Payam Kaufling	Shipping
14	123	Shanta Vollman	Shipping
15	124	Kevin Mourgog	Shipping
16	125	Julia Nayer	Shipping
17	126	Irene Mikkilineni	Shipping
18	127	James Landry	Shipping
19	128	Steven Markle	Shipping
20	129	Laura Bissot	Shipping

Report 2: Employee Name, Job Title, Salary (Employees with valid jobs only)

Worksheet		Query Builder	
1		SELECT	e.first_name ' ' e.last_name AS employee_name, j.job_title, e.salary
2		FROM	employees e
3		INNER JOIN	jobs j
4		ON	e.job_id = j.job_id;

Query Result		SQL Fetched 50 rows in 0.034 seconds	
	EMPLOYEE_NAME	JOB_TITLE	SALARY
1	William Gietz	Public Accountant	8300
2	Shelley Higgins	Accounting Manager	12008
3	Jennifer Whalen	Administration Assistant	4400
4	Steven King	President	24000
5	Lex De Haan	Administration Vice President	17000
6	Neena Kochhar	Administration Vice President	17000
7	John Chen	Accountant	8200
8	Daniel Faviet	Accountant	9000
9	Luis Popp	Accountant	6900
10	Ismael Sciarra	Accountant	7700
11	Jose Manuel Urman	Accountant	7800
12	Nancy Greenberg	Finance Manager	12008
13	Susan Mavris	Human Resources Representative	6500
14	Alexander Hunold	Programmer	9000
15	Bruce Ernst	Programmer	6000
16	Diana Lorentz	Programmer	4200
17	Valli Pataballa	Programmer	4800
18	David Austin	Programmer	4800
19	Michael Hartstein	Marketing Manager	13000
20	Pat Fay	Marketing Representative	6000

Report 3: Employee, Department, Department City

Worksheet		Query Builder
1		<code>SELECT e.first_name ' ' e.last_name AS employee_name, d.department_name, l.city</code>
2		<code>FROM employees e</code>
3		<code>INNER JOIN departments d</code>
4		<code>ON e.department_id = d.department_id</code>
5		<code>INNER JOIN locations l</code>
6		<code>ON d.location_id = l.location_id;</code>

Query Result x	
	SQL Fetched 50 rows in 0.044 seconds
EMPLOYEE_NAME	DEPARTMENT_NAME CITY
1 Ellen Abel	Sales Oxford
2 Sundar Ande	Sales Oxford
3 Mozhe Atkinson	Shipping South San Francisco
4 David Austin	IT Southlake
5 Hermann Baer	Public Relations Munich
6 Shelli Baida	Purchasing Seattle
7 Amit Banda	Sales Oxford
8 Elizabeth Bates	Sales Oxford
9 Sarah Bell	Shipping South San Francisco
10 David Bernstein	Sales Oxford
11 Laura Bissot	Shipping South San Francisco
12 Harrison Bloom	Sales Oxford
13 Alexis Bull	Shipping South San Francisco
14 Anthony Cabrio	Shipping South San Francisco
15 Gerald Cambrault	Sales Oxford
16 Nanette Cambrault	Sales Oxford
17 John Chen	Finance Seattle
18 Kelly Chung	Shipping South San Francisco

Report 4: Employee, Department, Country, Region

Worksheet

Query Builder

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SELECT

e.first_name || ' ' || e.last_name AS employee_name, d.department_name, c.country_name, r.region_name

FROM

employees e

INNER JOIN

departments d

ON

e.department_id = d.department_id

INNER JOIN

locations l

ON

d.location_id = l.location_id

INNER JOIN

countries c

ON

l.country_id = c.country_id

INNER JOIN

regions r

ON

c.region_id = r.region_id;

Query Result

x

SQL

| Fetched 50 rows in 0.003 seconds

	EMPLOYEE_NAME	DEPARTMENT_NAME	COUNTRY_NAME	REGION_NAME
1	Ellen Abel	Sales	United Kingdom	Europe
2	Sundar Ande	Sales	United Kingdom	Europe
3	Mozhe Atkinson	Shipping	United States of America	Americas
4	David Austin	IT	United States of America	Americas
5	Hermann Baer	Public Relations	Germany	Europe
6	Shelli Baida	Purchasing	United States of America	Americas
7	Amit Banda	Sales	United Kingdom	Europe
8	Elizabeth Bates	Sales	United Kingdom	Europe
9	Sarah Bell	Shipping	United States of America	Americas
10	David Bernstein	Sales	United Kingdom	Europe
11	Laura Bissot	Shipping	United States of America	Americas
12	Harrison Bloom	Sales	United Kingdom	Europe
13	Alexis Bull	Shipping	United States of America	Americas
14	Anthony Cabrio	Shipping	United States of America	Americas
15	Gerald Cambrault	Sales	United Kingdom	Europe

2. Organization Structure Analysis

Report 5: Complete Employee List (Including employees without departments)

Worksheet

Query Builder

1

SELECT

e.employee_id, e.first_name || ' ' || e.last_name AS employee_name, d.department_name

2

FROM

employees e

3

LEFT JOIN

departments d

4

ON

e.department_id = d.department_id;

Query Result

x

SQL

| Fetched 50 rows in 0.014 seconds

	EMPLOYEE_ID	EMPLOYEE_NAME	DEPARTMENT_NAME
1	200	Jennifer Whalen	Administration
2	201	Michael Hartstein	Marketing
3	202	Pat Fay	Marketing
4	114	Den Raphaely	Purchasing
5	115	Alexander Khoo	Purchasing
6	116	Shelli Baida	Purchasing
7	117	Sigal Tobias	Purchasing
8	118	Guy Himuro	Purchasing
9	119	Karen Colmenares	Purchasing
10	203	Susan Mavris	Human Resources
11	120	Matthew Weiss	Shipping
12	121	Adam Fripp	Shipping
13	122	Payam Kaufling	Shipping
14	123	Shanta Vollman	Shipping
15	124	Kevin Mourgog	Shipping
16	125	Julia Nayer	Shipping
17	126	Irene Mikkilineni	Shipping
18	127	James Landry	Shipping
19	128	Steven Markle	Shipping
20	129	Laura Bissot	Shipping

Report 6: Complete Department List (Including departments without employees)

Worksheet		Query Builder	
1		<code>SELECT</code>	<code>d.department_id, d.department_name, e.first_name ' ' e.last_name AS employee_name</code>
2		<code>FROM</code>	<code>departments d</code>
3		<code>LEFT JOIN</code>	<code>employees e</code>
4		<code>ON</code>	<code>d.department_id = e.department_id</code>
5		<code>ORDER BY</code>	<code>d.department_id;</code>
Query Result x			
SQL Fetched 50 rows in 0.007 seconds			
	DEPARTMENT_ID	DEPARTMENT_NAME	EMPLOYEE_NAME
1	10	Administration	Jennifer Whalen
2	20	Marketing	Pat Fay
3	20	Marketing	Michael Hartstein
4	30	Purchasing	Sigal Tobias
5	30	Purchasing	Karen Colmenares
6	30	Purchasing	Shelli Baida
7	30	Purchasing	Den Raphaely
8	30	Purchasing	Alexander Khoo
9	30	Purchasing	Guy Himuro
10	40	Human Resources	Susan Mavris
11	50	Shipping	Kevin Feeney
12	50	Shipping	Jean Fleaur
13	50	Shipping	Adam Fripp
14	50	Shipping	Timothy Gates
15	50	Shipping	Ki Gee
16	50	Shipping	Girard Geoni
17	50	Shipping	Douglas Grant
18	50	Shipping	Vance Jones
19	50	Shipping	Payam Kaufling

Report 7: Departments With No Employees

Worksheet

Query Builder

1

 SELECT d.department_id, d.department_name

2

FROM departments d

3

LEFT JOIN employees e

4

ON d.department_id = e.department_id

5

WHERE e.employee_id IS NULL;

 Query Result x

    SQL | All Rows Fetched: 16 in 0.007 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME
1	120	Treasury
2	130	Corporate Tax
3	140	Control And Credit
4	150	Shareholder Services
5	160	Benefits
6	170	Manufacturing
7	180	Construction
8	190	Contracting
9	200	Operations
10	210	IT Support
11	220	NOC
12	230	IT Helpdesk
13	240	Government Sales
14	250	Retail Sales
15	260	Recruiting
16	270	Payroll

3. Management Hierarchy Analysis

Report 8: Employee Name and Direct Manager Name (Self Join)

Worksheet

Query Builder

1

Report 9: Employees Without Managers

Worksheet		Query Builder	
1		SELECT	employee_id, first_name ' ' last_name AS employee_name
2		FROM	employees
3		WHERE	manager_id IS NULL;

Query Result x	
SQL All Rows Fetched: 1 in 0.005 seconds	
EMPLOYEE_ID	EMPLOYEE_NAME
1	100 Steven King

These employees are at the top of the organizational hierarchy (such as the CEO or executive leadership) and therefore do not report to another employee.

4. Advanced Business Analysis

Report 10: Employees Earning More Than Their Managers

Worksheet	Query Builder
1	<code>SELECT e.first_name ' ' e.last_name AS employee_name,</code>
2	<code> e.salary AS employee_salary,</code>
3	<code> m.first_name ' ' m.last_name AS manager_name,</code>
4	<code> m.salary AS manager_salary</code>
5	<code>FROM employees e</code>
6	<code>INNER JOIN employees m</code>
7	<code>ON e.manager_id = m.employee_id</code>
8	<code>WHERE e.salary > m.salary;</code>

Query Result x			
All Rows Fetched: 2 in 0.006 seconds			
EMPLOYEE_NAME	EMPLOYEE_SALARY	MANAGER_NAME	MANAGER_SALARY
1 Lisa Ozer	11500	Gerald Cambrault	11000
2 Ellen Abel	11000	Eleni Zlotkey	10500

Report 11: Departments Where Department Manager Works in Same Department

Worksheet	Query Builder
1	<code>SELECT d.department_id, d.department_name, e.first_name ' ' e.last_name AS manager_name</code>
2	<code>FROM departments d</code>
3	<code>INNER JOIN employees e</code>
4	<code>ON d.manager_id = e.employee_id</code>
5	<code>WHERE d.department_id = e.department_id;</code>

Query Result x		
All Rows Fetched: 11 in 0.028 seconds		
DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_NAME
1	90 Executive	Steven King
2	60 IT	Alexander Hunold
3	100 Finance	Nancy Greenberg
4	30 Purchasing	Den Raphaely
5	50 Shipping	Adam Fripp
6	80 Sales	John Russell
7	10 Administration	Jennifer Whalen
8	20 Marketing	Michael Hartstein
9	40 Human Resources	Susan Mavris
10	70 Public Relations	Hermann Baer
11	110 Accounting	Shelley Higgins

5. Executive Dashboard Reports

Dashboard 1: Employee, Job Title, Department, City, Country

Worksheet

Query Builder

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SELECT e.first_name || ' ' || e.last_name AS employee_name, j.job_title, d.department_name, l.city, c.country_name

FROM employees e

INNER JOIN jobs j

ON e.job_id = j.job_id

LEFT JOIN departments d

ON e.department_id = d.department_id

LEFT JOIN locations l

ON d.location_id = l.location_id

LEFT JOIN countries c

ON l.country_id = c.country_id;

Query Result x

SQL | Fetched 50 rows in 0.01 seconds

EMPLOYEE_NAME	JOB_TITLE	DEPARTMENT_NAME	CITY	COUNTRY_NAME
1 Jennifer Whalen	Administration Assistant	Administration	Seattle	United States of America
2 Michael Hartstein	Marketing Manager	Marketing	Toronto	Canada
3 Pat Fay	Marketing Representative	Marketing	Toronto	Canada
4 Shelli Baida	Purchasing Clerk	Purchasing	Seattle	United States of America
5 Alexander Khoo	Purchasing Clerk	Purchasing	Seattle	United States of America
6 Sigal Tobias	Purchasing Clerk	Purchasing	Seattle	United States of America
7 Guy Himuro	Purchasing Clerk	Purchasing	Seattle	United States of America
8 Karen Colmenares	Purchasing Clerk	Purchasing	Seattle	United States of America
9 Den Raphaely	Purchasing Manager	Purchasing	Seattle	United States of America
10 Susan Mavris	Human Resources Representative	Human Resources	London	United Kingdom
11 Nandita Sarchand	Shipping Clerk	Shipping	South San Francisco	United States of America
12 Alexis Bull	Shipping Clerk	Shipping	South San Francisco	United States of America
13 Julia Dellinger	Shipping Clerk	Shipping	South San Francisco	United States of America
14 Anthony Cabrio	Shipping Clerk	Shipping	South San Francisco	United States of America

Dashboard 2: Department, Manager, City, Country (Including departments without managers)

Worksheet

Query Builder

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SELECT

d.department_name, m.first_name || ' ' || m.last_name AS manager_name, l.city, c.country_name

FROM

departments d

LEFT JOIN

employees m

ON

d.manager_id = m.employee_id

LEFT JOIN

locations l

ON

d.location_id = l.location_id

LEFT JOIN

countries c

ON

l.country_id = c.country_id;

Query Result x

SQL | All Rows Fetched: 27 in 0.014 seconds

	DEPARTMENT_NAME	MANAGER_NAME	CITY	COUNTRY_NAME
1	Executive	Steven King	Seattle	United States of America
2	IT	Alexander Hunold	Southlake	United States of America
3	Finance	Nancy Greenberg	Seattle	United States of America
4	Purchasing	Den Raphaely	Seattle	United States of America
5	Shipping	Adam Fripp	South San Francisco	United States of America
6	Sales	John Russell	Oxford	United Kingdom
7	Administration	Jennifer Whalen	Seattle	United States of America
8	Marketing	Michael Hartstein	Toronto	Canada
9	Human Resources	Susan Mavris	London	United Kingdom
10	Public Relations	Hermann Baer	Munich	Germany
11	Accounting	Shelley Higgins	Seattle	United States of America
12	Treasury		Seattle	United States of America
13	Corporate Tax		Seattle	United States of America
14	Control And Credit		Seattle	United States of America
15	Shareholder Services		Seattle	United States of America
16	Benefits		Seattle	United States of America

Dashboard 3: Employees Working in Same Department as Their Manager

Worksheet		Query Builder	
1	SELECT	e.first_name ' ' e.last_name AS employee_name,	
2		m.first_name ' ' m.last_name AS manager_name,	
3		d.department_name	
4	FROM	employees e	
5	INNER JOIN	employees m	
6	ON	e.manager_id = m.employee_id	
7	INNER JOIN	departments d	
8	ON	e.department_id = d.department_id	
9	WHERE	e.department_id = m.department_id;	

Query Result		SQL Fetched 50 rows in 0.005 seconds	
	EMPLOYEE_NAME	MANAGER_NAME	DEPARTMENT_NAME
1	Neena Kochhar	Steven King	Executive
2	Lex De Haan	Steven King	Executive
3	Bruce Ernst	Alexander Hunold	IT
4	David Austin	Alexander Hunold	IT
5	Valli Pataballa	Alexander Hunold	IT
6	Diana Lorentz	Alexander Hunold	IT
7	Daniel Faviet	Nancy Greenberg	Finance
8	John Chen	Nancy Greenberg	Finance
9	Ismael Sciarra	Nancy Greenberg	Finance
10	Jose Manuel Urman	Nancy Greenberg	Finance
11	Luis Popp	Nancy Greenberg	Finance
12	Alexander Khoo	Den Raphaely	Purchasing
13	Shelli Baida	Den Raphaely	Purchasing
14	Sigal Tobias	Den Raphaely	Purchasing
15	Guy Himuro	Den Raphaely	Purchasing

Dashboard 4: Employees Whose Managers Work in Different Departments

Worksheet

Query Builder

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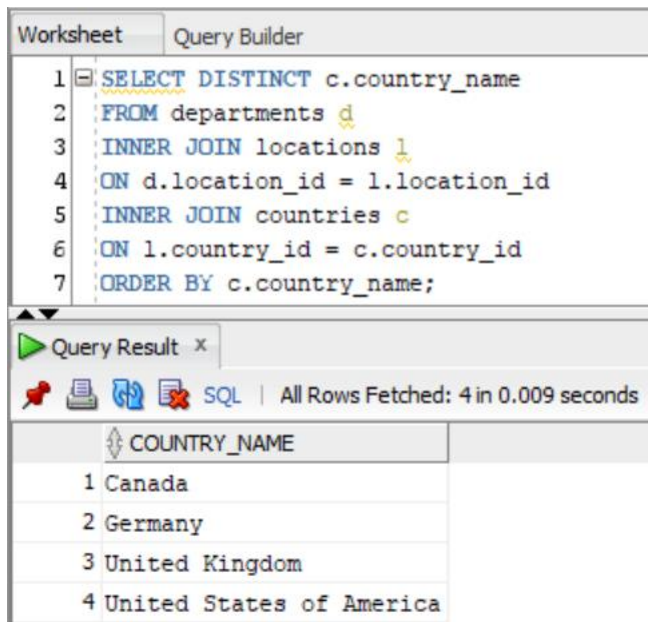
5

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Dashboard 5: Distinct Countries Containing Company Departments



The screenshot displays a database query builder interface. At the top, there are two tabs: 'Worksheet' and 'Query Builder'. The 'Query Builder' tab is active, showing an SQL query in a text area. The query is as follows:

```
1 SELECT DISTINCT c.country_name
2 FROM departments d
3 INNER JOIN locations l
4 ON d.location_id = l.location_id
5 INNER JOIN countries c
6 ON l.country_id = c.country_id
7 ORDER BY c.country_name;
```

Below the query editor, there is a 'Query Result' section. It includes a green play button icon, a close button (x), and a status bar that reads 'All Rows Fetched: 4 in 0.009 seconds'. Below this, there is a table with the following data:

	COUNTRY_NAME
1	Canada
2	Germany
3	United Kingdom
4	United States of America

JOINS Used:

1. INNER JOIN → Matching records only.
2. LEFT JOIN → Keep all rows from the left table.
3. SELF JOIN → EMPLOYEES joined with themselves for manager relationships.